



What is Multithreading?

Multithreading means:

👉 Running **multiple tasks at the same time** inside one program.

Example:

- Downloading a file while printing progress
 - Playing music while using an app
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◆ What is a Thread?

A **thread** is a small unit of a program.

A program can have:

- 1 thread → Single task
 - Multiple threads → Multiple tasks
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◆ Why Use Multithreading?

- ✓ Makes program faster (for I/O work)
 - ✓ Runs background tasks
 - ✓ Better performance for file, network, waiting tasks
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Threading Module in Python

First import:

```
import threading
```

✓ Example 1: Create a Simple Thread

```
import threading
```

```
def task():  
    print("Thread is running")
```

```
t1 = threading.Thread(target=task)  
t1.start()
```

Explanation:

- `Thread()` → creates thread
- `target=task` → function to run
- `start()` → starts thread

✓ Example 2: Main Thread + Child Thread

```
import threading
```

```
def task():  
    print("Child thread")
```

```
t1 = threading.Thread(target=task)  
t1.start()
```

```
print("Main thread")
```

Both run together.

Example 3: Using join()

`join()` makes main program wait.

```
import threading
```

```
def task():
```

```
    print("Thread running")
```

```
t1 = threading.Thread(target=task)
```

```
t1.start()
```

```
t1.join()
```

```
print("Program finished")
```

Example 4: Multiple Threads

```
import threading
```

```
import time
```

```
def print_numbers():
```

```
    for i in range(3):
```

```
print(i)

time.sleep(1)
```

```
def print_letters():

    for ch in ['A','B','C']:

        print(ch)

        time.sleep(1)
```

```
t1 = threading.Thread(target=print_numbers)

t2 = threading.Thread(target=print_letters)
```

```
t1.start()
```

```
t2.start()
```

Output will mix numbers and letters.



What is Lock? (Basic Idea)

If two threads use the same variable, problem can happen.

To prevent this, use **Lock**.

```
import threading
```

```
lock = threading.Lock()
```

```
def task():
```

```
    with lock:
```

```
print("Safe section")
```

Lock means:

👉 Only one thread can enter at a time.

Important Basic Functions

Function	Meaning
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<code>Thread()</code>	Create thread
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<code>start()</code>	Start thread
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<code>join()</code>	Wait for thread
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<code>Lock()</code>	Prevent conflict
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Important Note (Very Simple)

Python has something called **GIL**:

👉 Only one thread runs at a time for CPU work.

So:

- Good for file/network tasks 
 - Not good for heavy calculations 
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Simple Summary

- Multithreading = Multiple tasks at same time
- Use `threading` module
- Use `start()` to run thread
- Use `join()` to wait
- Use `Lock()` to avoid problems

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